Advanced AI Dissertation

Quantum Ensemble Stacking with State Preservation

Edward Mallia

Student ID: 25233259

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Supervisor: Dr Simon Caton

UCD School of Computer Science

University College Dublin

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Abstract

- Context: practical limitations of large variational quantum classifiers and the motivation for ensembles of small circuits.
- Problem: conventional stacking measures each quantum base learner before aggregation, potentially creating an information bottleneck.
- Aim: design and evaluate a progression from measured ensemble aggregation to classical stacking, quantum re-encoding and measurement-free coherent stacking.
- Method: configurable VQC base learners, ECOC decomposition, systematic VQC architecture selection, controlled aggregation comparisons and multiclass tabular datasets.
- Evaluation: accuracy, macro-F1, class-level behaviour, convergence, training time and circuit-resource requirements.
- Principal findings: insert only after the final experiments and analysis are complete.
- Final conclusion: state whether coherent stacking was feasible, beneficial and practically justified relative to measured baselines.

Chapter 1: Introduction

1.1 Background and Motivation

Key points to cover.

- Introduce QML and variational quantum models as a near-term approach to supervised learning.
- Explain the practical restrictions of current quantum models: qubit limits, circuit depth, noise, expensive simulation and difficult optimisation.
- Motivate ensembles of several small VQCs as an alternative to one large monolithic circuit.
- Introduce stacking as a learned aggregation mechanism rather than fixed voting or averaging.
- Establish the specifically quantum issue: intermediate measurement converts a quantum state into classical outputs before the meta-learner can act on it.
- Motivate the central investigation without assuming that unmeasured states necessarily improve classification.
- End by narrowing the discussion from general QML to measurement-free quantum stacking.

Research notes and evidence.

- Ensemble methods combine complementary information from multiple learners and can improve accuracy, stability and generalisation on complex, noisy, imbalanced or high-dimensional data [7].
- Stacking learns a second-level combiner from base-model predictions, making it a suitable framework for comparing classical and quantum aggregation mechanisms [6].
- Quantum ensembles can retain the contributions of multiple classifiers rather than selecting one final model, motivating ensembles of simple quantum learners [23].
- Small or shallow quantum learners have been proposed as a modular response to NISQ-era limits in circuit depth, trainability and noise [9, 15, 10].
- Classical ensembles of small quantum classifiers have improved measured performance in several simulated and hardware-oriented studies, supporting ensemble construction as a relevant QML direction [25, 13, 27].
- QML remains limited by data loading, output extraction, benchmarking, noise and trainability, so the dissertation should avoid broad or unsupported quantum-advantage claims [4, 8].

1.2 Problem Statement

Key points to cover.

- Define conventional measured quantum stacking: independently measure each VQC, concatenate its classical outputs and train a meta-learner.
- Explain that a quantum meta-learner operating on re-encoded probabilities remains downstream of the same measurement boundary.
- Define the proposed alternative: preserve each trained base circuit as an unmeasured subcircuit and append a trainable meta-circuit before final measurement.
- State the uncertainty being tested: whether the preserved state contains exploitable information that is useful for classification.
- Identify the competing practical cost: wider and deeper circuits, more two-qubit gates, harder optimisation and increased simulation or hardware demands.
- Frame the dissertation as a controlled empirical comparison rather than a proof that measurement necessarily harms ensemble performance.

Research notes and evidence.

- An n-qubit state is described by 2^n amplitudes, while measurement yields classical outcomes and does not directly reveal the full state description [20].
- Entanglement stores information in relationships between qubits, motivating investigation of whether cross-register operations can exploit relationships unavailable after independent base measurements [20].
- Quantum states are fragile and cannot be freely copied or persistently stored, making quantum intermediate representations fundamentally different from saved classical stacking features [20].
- Increased circuit size and global entanglement can worsen optimisation, noise exposure and barren-plateau behaviour [8, 5].
- Existing work does not provide a direct controlled comparison of classical measured stacking, quantum re-encoding and a coherent meta-circuit applied to trained VQC base states.

1.3 Aim

Key points to cover.

- State one overall aim: to design, implement and evaluate a quantum ensemble architecture whose aggregation stage progresses from measured methods to a fully quantum, measurement-free stacking condition.
- Emphasise that the base VQCs and ECOC decomposition provide a controlled foundation; the primary research focus is the stacking boundary and meta-level aggregation.

1.4 Research Objectives

1. Develop configurable and trainable VQC base learners for small multiclass tabular datasets.
2. Investigate VQC architecture choices and select a compact set of feasible base-learner designs.
3. Construct a controlled ECOC ensemble of simple quantum learners.
4. Establish direct measured ECOC aggregation as the basic quantum ensemble baseline.
5. Implement classical measured stacking using stratified out-of-fold meta-training.
6. Implement a quantum meta-learner operating on re-encoded measured base outputs.
7. Develop a coherent architecture that removes intermediate measurements and appends a trainable quantum meta-circuit to frozen base circuits.
8. Compare predictive performance, optimisation behaviour, computational cost and circuit-resource requirements across aggregation conditions.
9. Analyse which dataset and ensemble characteristics are associated with any observed stacking benefit.

1.5 Research Questions

1. Can an ensemble of small VQC base learners provide a viable foundation for multiclass quantum classification?
2. How does classical stacking of measured VQC outputs compare with direct measured ECOC aggregation?
3. Does a quantum meta-learner trained on re-encoded measured outputs provide an advantage over a classical meta-learner?
4. Can trained VQC base learners be composed into a trainable measurement-free quantum stacking circuit?
5. Does preserving the base learners' quantum states improve classification performance, macro-F1, robustness or convergence relative to measured stacking?
6. What additional computational and quantum-resource costs are introduced by measurement-free stacking?
7. For which dataset or ensemble characteristics, if any, is coherent quantum stacking most beneficial?

1.6 Expected and Actual Contributions

Key points to cover.

- A configurable framework for VQC architecture construction and evaluation.
- A systematic VQC architecture-search pipeline and cross-dataset ranking process.
- A controlled ECOC framework for generating simple multiclass quantum base learners.
- A strong out-of-fold classical stacking baseline for measured VQC ensembles.
- A quantum re-encoding comparison that isolates the effect of the meta-learner type.
- A measurement-free architecture formed by composing frozen VQC base circuits with a trainable coherent meta-circuit.
- An empirical analysis of measurement as a possible base-to-meta information bottleneck.
- A performance–cost analysis covering prediction quality, training cost and circuit resources.
- Rewrite this list near submission so that it states only contributions that were actually completed and evaluated.

1.7 Scope and Delimitations

Key points to cover.

- The central contribution concerns aggregation, not a novel ECOC coding theory or a general-purpose new VQC.
- The study focuses on supervised multiclass classification using small and moderate tabular datasets.
- ECOC is used as practical scaffolding for repeatable low-complexity base tasks.
- Base learners are deliberately small so that the stacking mechanism remains the main experimental variable.
- Most experiments are expected to rely on simulation because coherent statevector cost scales exponentially with total qubit count.
- Include noisy simulation or IBM hardware only if those experiments are actually completed.
- Do not claim quantum speed-up or quantum advantage unless directly supported by the final evidence.

1.8 Dissertation Structure

Key points to cover.

- Summarise Chapters 2–6 in one short paragraph after the chapters have stabilised.

Chapter 2: Related Work

2.1 Ensemble Learning

2.1.1 Motivation for Ensemble Learning

Key points to cover.

- Explain why multiple imperfect learners can outperform or stabilise a single learner.
- Introduce the accuracy–diversity trade-off.
- Identify sources of diversity: data samples, feature subsets, model architectures, class decompositions, initialisation and prediction behaviour.
- Relate ensemble diversity to the dissertation's feature subspacing, bagging, ECOC tasks and VQC-template variation.

Research notes and evidence.

- Ensemble diversity can be introduced through sample subsets, feature subspaces, classifier selection, weighting and ensemble-level optimisation [7].
- Ensemble quality depends on the joint behaviour of accuracy, diversity, generalisation, stability, sparsity and complexity rather than constituent accuracy alone [7].
- Random-subspace methods provide direct precedent for assigning different feature subsets to VQC base learners [7, 17].
- Prior VQC composition work identifies prediction disagreement as a stronger indicator of ensemble contribution than individual F1 or architectural difference, supporting analysis of actual base-prediction diversity [16].
- Diversity-aware QBoost work explicitly optimises disagreement alongside prediction loss and regularisation, providing precedent for treating diversity as a design objective rather than only a post-hoc statistic [12].

2.1.2 Bagging, Boosting and Stacking

Key points to cover.

- Distinguish parallel bagging, sequential boosting and two-level stacking.
- Explain why stacking is the principal aggregation method in this dissertation.
- Describe fixed aggregation, weighted aggregation and learned aggregation.
- Clarify that bagging is also used within the project as a source of base-learner diversity rather than as the main research contribution.

Research notes and evidence.

- Classical supervised ensembles commonly include bagging, boosting, random forests, random subspaces and stacking [7].
- Stacking uses base predictions as second-level input features and trains a combiner to learn how those predictions should be integrated [6].
- Different stackers can favour different class-level behaviours, showing that the meta-learner is a substantive design choice rather than neutral plumbing [6].
- Prior VQC ensemble work reports stronger gains from stacking than from bagging, supporting the dissertation's emphasis on learned aggregation [16].

2.1.3 Out-of-Fold Stacking

Key points to cover.

- Explain why training the meta-learner on in-sample base predictions creates leakage and overly optimistic performance.
- Define stratified K-fold out-of-fold prediction generation.
- Explain the distinction between temporary fold ensembles and the final base ensemble trained on the full training partition.
- Connect this directly to the implemented `StackedECOC` training procedure.

Research notes and evidence.

- Five-fold stacking predictions are used in prior hybrid ensemble work to create valid learned features before downstream quantum classification [29].
- Modified stacking uses feature-subspace base models and validation-driven model selection before training the combiner, providing precedent for separating base-model generation from meta-training [6].

2.1.4 Error-Correcting Output Codes

Key points to cover.

- Introduce ECOC as a method for decomposing multiclass classification into several simpler learner tasks.
- Define the coding matrix, class codewords, learner columns or rows according to the implementation convention, and decoding.
- Explain binary ECOC and the optional generalisation to greater decomposition depth.
- Explain why ECOC is appropriate here: controllable learner count, simple VQC tasks and consistent use across datasets.
- State explicitly that ECOC is not the central novelty.

Research notes and evidence.

- ECOC decomposes a c-class problem into several simpler learner problems and decodes the resulting prediction codeword against class codewords [14].
- Strong ECOC matrices aim to maintain separation between class codewords while avoiding redundant or constant learners [14].
- N-ary ECOC generalises binary class partitions but increases the complexity of each base task, supporting binary decomposition as the principal dissertation setting [14].
- Classical ECOC optimisation literature does not address VQC trainability, qubit budgets or coherent quantum stacking [14].

2.2 Quantum Machine Learning

2.2.1 Near-Term Quantum Machine Learning

Key points to cover.

- Distinguish fault-tolerant quantum algorithms from NISQ-compatible hybrid variational approaches.
- Introduce the hybrid quantum–classical optimisation loop.
- Explain why VQCs are appropriate to the dissertation's implementation and resource constraints.
- Treat quantum computing as a specialised accelerator or experimental computational model rather than a wholesale replacement for classical ML.

Research notes and evidence.

- Fault-tolerant QML may support stronger theoretical speed-ups but requires error-corrected devices, whereas QNNs and kernels are more experimentally accessible in the NISQ setting [8].
- Near-term variational algorithms repeatedly execute a parameterised quantum circuit, measure its outputs and update parameters through a classical optimiser [20, 8].
- Input preparation, output reconstruction and practical resource estimates can undermine theoretical QML speed-up claims [4].
- Early QML literature distinguishes quantum acceleration of classical subroutines from genuinely quantum learning models, supporting the dissertation's focus on the learning and aggregation mechanism itself [24].

2.2.2 Variational Quantum Classifiers

Key points to cover.

- Define a VQC as feature encoding, parameterised quantum transformations, entangling operations, measurement and classical optimisation.
- Describe how measurements are converted into class probabilities in this project.
- Introduce local versus fuller measurement choices.
- Explain why architecture selection matters for expressivity, trainability and generalisation.

Research notes and evidence.

- A standard QNN workflow consists of encoding, trainable quantum processing, measurement, loss evaluation and classical parameter updates [8].
- Circuit design is governed by the interaction between expressivity, trainability and generalisation [8].
- VQCs can be sensitive to initialisation, dataset size, circuit depth and cost-function concentration [28, 9].

2.2.3 Quantum Feature Encoding and Re-uploading

Key points to cover.

- Introduce angle encoding and the mapping of classical features to rotation gates.
- Explain feature density, feature assignment across qubits and repeated data re-uploading.
- Discuss the trade-off between richer feature representation and increased depth or gate count.
- Introduce pairwise feature interactions through RZZ-based encoding where relevant.

Research notes and evidence.

- Data re-uploading is a recognised strategy for increasing the expressivity of classical-data quantum feature maps [8].
- Repeated feature-encoding rotations can increase accessible Fourier frequencies and classifier expressivity [13].
- Feature-map ordering and the placement of trainable layers can materially change trainability and eliminate redundant gates [22].
- Higher feature dimensionality increases circuit width or depth and can make full-feature VQC training prohibitive, motivating feature subspacing [17].

2.2.4 Ansatz Design and Entanglement

Key points to cover.

- Describe parameterised rotation layers and entangling layers.
- Discuss CZ and RZZ entanglers, linear/ring/odd–even connectivity and hardware-aware design.
- Explain why the dissertation searches several qualitatively different shallow ansatz structures rather than committing to one arbitrary template.
- Relate the selected gate family to IBM Heron-compatible gates where appropriate.

Research notes and evidence.

- Similar-depth ansatz families can differ substantially in performance, creating an empirical "ansatz lottery" if only one architecture is tested [16].
- Shallow circuit families, restricted entanglement and hardware-efficient templates can improve practical trainability and reduce noise exposure [9, 15].
- Limited hardware connectivity may introduce SWAP operations, increasing depth, latency and error exposure [20].
- Several studies observe diminishing performance returns after a small number of trainable layers, supporting shallow initial designs [22, 10].

2.2.5 Trainability and Scaling

Key points to cover.

- Introduce barren plateaus, vanishing gradients, local minima and initialisation sensitivity.
- Explain how circuit width, depth, global measurements and entanglement can affect optimisation.
- Motivate shallow base learners, early stopping and staged training.
- Explain the exponential classical statevector cost of increasing the total coherent qubit count.

Research notes and evidence.

- Barren plateaus can cause gradient magnitudes to decrease exponentially with system size [8, 5].
- Initialisation, local cost functions, structured circuits and regularisation are common mitigation strategies [8].
- Partitioning one deep quantum layer into multiple shallow components can increase gradient variance and improve convergence in some settings [9].
- More global entanglement may worsen trainability and generalisation, while local or modular entanglement can reduce optimisation difficulty [15].

2.2.6 Measurement and Quantum Information

Key points to cover.

- Explain measurement as the final conversion from quantum state to classical outcome probabilities.
- Distinguish the mathematical state from the limited statistics obtained through measurement.
- Explain why independently measuring base registers prevents later coherent cross-register operations.
- Avoid claiming that all discarded state information is useful for the target task.
- Establish measurement as the empirical variable examined at the base-to-meta boundary.

Research notes and evidence.

- Amplitudes are not directly observable and expectation values require repeated measurement or exact simulation [20, 27].
- Measured ensemble pipelines are sensitive to shot noise and readout precision [27, 26].
- The no-cloning theorem and the absence of practical persistent quantum memory prevent treating quantum intermediate states like ordinary stored classical features [20].

2.3 Quantum Ensemble Learning

2.3.1 Coherent Superposition Ensembles

Key points to cover.

- Review early definitions of quantum ensembles based on superpositions over classifier parameters or indexed internal classifiers.
- Explain the theoretical appeal of parallel evaluation and amplitude-weighted aggregation.
- Distinguish these methods from the dissertation's finite set of independently trained VQC subcircuits.

Research notes and evidence.

- Schuld and Petruccione prepare a weighted superposition over classifier parameter states and extract a collective decision through final measurement [23].
- Araujo and da Silva compare untrained and trained classifier ensembles and find that optimisation is important on benchmark datasets [3].

- Weighted indexed ensembles can generate different internal data compositions in superposition and re-encode classically learned aggregation weights for coherent test-time execution [26].
- These approaches rely on idealised state preparation, indexed data access, post-selection or simulated amplitudes, limiting direct equivalence to near-term trained VQC stacks [23, 3, 26].

2.3.2 Measured Quantum Bagging and Boosting

Key points to cover.

- Review measured quantum base learners combined through voting, averaging, bagging or boosting.
- Explain what these approaches demonstrate about ensemble robustness and small-circuit modularity.
- Identify that they still measure base learners before aggregation.

Research notes and evidence.

- AdaBoost.Q uses measured class probabilities as confidence information for reweighting samples and combining quantum classifiers [25].
- Bagging and boosting of compact QNN or single-qubit classifiers can improve predictive performance while reducing matched training or circuit costs [13, 10].
- Bagging–QSVC and related measured quantum ensembles aggregate independently measured predictions classically [1].
- Quantum bootstrapped bagging combines quantum-inspired subsampling or clustering with measured or classical aggregation, providing robustness evidence but not coherent VQC meta-learning [18].
- These methods provide evidence that ensemble construction can help quantum classifiers, but they do not preserve base states for a coherent meta-circuit.

2.3.3 Measured and Hybrid Quantum Stacking

Key points to cover.

- Review stacking systems that use quantum learners as measured base models and a classical stacker.
- Review systems that combine classical and quantum learners in a classical stacking architecture.
- Identify their relevance as measured baselines rather than coherent architectures.

Research notes and evidence.

- Vasan et al. combine measured QSVC, QVC and QNN posterior probabilities with a Random Forest meta-learner [28].
- Kang and Shin combine measured ML, DNN and QNN predictions through a classical meta-model [11].
- Xiong et al. first use classical stacking for feature reduction and then apply a heterogeneous measured quantum voting ensemble [29].
- These studies report improved aggregate performance or minority-class behaviour, supporting measured stacking as a substantive baseline [28, 11, 29].
- None trains a coherent quantum meta-circuit directly on the unmeasured states of trained VQC base learners.

2.3.4 Quantum Meta-Learning after Measurement

Key points to cover.

- Review systems that re-encode classical base predictions into a quantum kernel or variational meta-model.
- Explain that they test the value of a quantum meta-model but not preservation of the original base states.
- Use this distinction to motivate the dissertation's re-encoding comparison condition.

Research notes and evidence.

- SEQSVM concatenates classical ensemble predictions into a compact meta-feature vector and feeds it to a quantum-kernel SVM [2].
- Quantum stacking studies can provide the quantum meta-learner with predicted classes, confidence values or reduced high-level features [27, 2].
- Re-encoding can introduce quantum processing at the meta-level, but cannot reconstruct information removed by the preceding base measurement.

2.3.5 Modular and Coherent Quantum Aggregation

Key points to cover.

- Review shallow ensemble layers, multi-chip VQCs, controlled-trajectory ensembles and amplitude-amplification aggregation.
- Separate approaches that preserve coherence from those that use modular circuits but aggregate classically.
- Explain where the proposed frozen-base coherent stack is similar and where it differs.

Research notes and evidence.

- Friedrich and Maziero replace a deeper quantum layer with several independently measured shallow circuits whose outputs are summed [9].
- Multi-chip ensemble VQCs partition inputs across small circuits but intentionally avoid cross-chip entanglement and combine measured outputs through a classical function [15].
- Rathi and Kumar create multiple transformed trajectories in superposition before a shared classifier and final ensemble measurement [19].
- Safina uses amplitude amplification to estimate aggregate class probability from classically prepared classifier probabilities [21].
- These methods do not implement the same staged process of independently training VQC base learners, freezing their parameters and appending a trainable cross-register meta-VQC.

2.4 Gap Analysis

2.4.1 Limitations of Existing Work

Key points to cover.

- Most practical quantum ensembles measure each component before aggregation.
- Hybrid and re-encoded quantum stackers operate only on classical outputs produced after collapse.
- Coherent ensemble proposals often rely on fault-tolerant assumptions, indexed data access, idealised state preparation or non-VQC classifier models.
- Modular VQC studies often avoid cross-register coupling for scalability and trainability.
- Existing studies rarely hold the base ensemble fixed while comparing direct measured aggregation, classical stacking, quantum re-encoding and coherent stacking.
- Many evaluations use small binary datasets or limited simulator settings and do not report both predictive and circuit-resource costs.

Research notes and evidence.

- Early quantum ensemble work explicitly leaves open alternative coherent definitions of quantum ensembles [23].
- Measured quantum stacking studies do not investigate direct aggregation of unmeasured trained VQC states [28, 11].
- Quantum meta-learning after re-encoding leaves the measurement boundary intact [2, 27].
- Shallow or multi-chip quantum ensembles improve modularity but still sum or classically combine measured outputs [9, 15].

- Quantum bagging frameworks may assume QRAM-style access or retain classical clustering and aggregation steps, leaving the trained coherent VQC stacking problem unresolved [18].
- Supervisor-provided VQC ensemble studies identify VQC meta-learning as a natural extension but do not test coherent measurement-free stacking [17, 16].

2.4.2 Research Gap Addressed by This Dissertation

Key points to cover.

- State the gap in one precise paragraph: limited empirical evidence exists for stacking trained VQC base learners by preserving their unmeasured states and applying a trainable quantum meta-circuit across their registers.
- Explain that the study responds through a controlled sequence of aggregation conditions sharing the same basic ensemble foundation.
- Emphasise that the comparison evaluates both prediction quality and the additional cost of state preservation.

Chapter 3: Approach and Methodology

3.1 Research Design

Key points to cover.

- Describe the dissertation as a controlled empirical comparison of increasingly quantum aggregation mechanisms.
- Define the principal independent variable as aggregation strategy.
- Define dependent measures: predictive metrics, optimisation behaviour, runtime and quantum resources.
- Explain which factors are held consistent across conditions: datasets, preprocessing, outer splits, base-learner pool and evaluation procedure.
- Explain where exact equivalence is impossible, such as classical versus quantum meta-model capacity.

3.1.1 Experimental Conditions

1. Classical reference models: SVC and Random Forest.
2. Selected individual VQCs.
3. Direct measured QuantumECOC aggregation.
4. Classical measured StackedECOC using out-of-fold base outputs.
5. Quantum meta-learning through re-encoded measured outputs.
6. Measurement-free coherent stacking using frozen base circuits and a trainable quantum meta-circuit.

Research notes and evidence.

- The comparison should isolate the effect of aggregation rather than allowing every condition to use unrelated base learners or preprocessing.
- Prior ensemble design studies support varying feature subsets, learner selection and ensemble size in a bounded structured search rather than an unrestricted sweep [7, 6, 16].

3.2 System Overview

Key points to cover.

- Add an architecture figure showing: dataset → preprocessing → ECOC decomposition → VQC base learners → selected aggregation condition → final prediction.
- Separate the system into a base-ensemble level and a stacking/meta-learning level.
- Add a second figure contrasting the information flow in measured, re-encoded and coherent stacking.
- Explain the incremental implementation path from direct ensemble aggregation to the final coherent model.

3.3 Implementation and Computational Environment

Key points to cover.

- Python version and core libraries: PyTorch, PennyLane, NumPy, pandas and scikit-learn.
- Custom VQC layer construction, forward pass, training/validation loop, prediction functions and result collection.
- Configurable ansatz definitions and template serialisation.
- Simulation devices and differentiation methods actually used.
- UCD SONIC HPC environment, Slurm execution and multiprocessing where relevant.
- Structured JSONL result logging, deterministic job identifiers and completed-job skipping.
- GitHub repository and reproducibility information.
- Include Qiskit or IBM Runtime only for work that is actually used in the final experiments.

Research notes and evidence.

- PennyLane, PyTorch and Qiskit are established tools for hybrid quantum-classical experimentation [8, 20].
- Simulator results represent best-case or modelled behaviour unless an explicit device-noise model or hardware execution is included [27].

3.4 Datasets

3.4.1 Dataset Roles

Key points to cover.

- Define sanity-check datasets used to validate implementation and basic trainability.

- Define main evaluation datasets used for systematic comparisons.
- Define final stress-test datasets used on a reduced set of selected configurations.
- State the final list only after the full experiment plan is frozen.

3.4.2 Dataset Characteristics

Table 3.1: Dataset characteristics and experimental roles.

Dataset	Samples	Features	Classes	Imbalance	Feature types	Role

Key points to cover.

- Discuss sample size, feature dimensionality, number of classes, imbalance, missing values and mixed data types.
- Explain why the set supports analysis across different levels of classification difficulty.
- Record any original unconventional train/test splits and justify whether they were retained or replaced.

3.5 Data Preprocessing

3.5.1 Cleaning and Missing Values

- Dataset-specific invalid-value handling.
- Numerical imputation strategy and fitted values.
- Categorical missing-value handling.

3.5.2 Categorical Encoding

- Encoding strategy for nominal and ordinal variables.
- Avoidance of train–test leakage.
- Preservation of final feature mappings for reproducibility.

3.5.3 Scaling

- Scaling range required by angle encoding, including the implemented $[0, \pi]$ range where applicable.
- Fit the scaler on training data and apply it to validation/test data.
- Explain any condition-specific scaling required by classical or quantum models.

3.5.4 Data Splits

- Outer train/test split and stratification.
- Validation data used for early stopping or base-learner weighting.
- Inner out-of-fold splits used only for meta-feature construction.
- Repeated seeds or runs, where computationally feasible.

3.6 Variational Quantum Classifier

3.6.1 VQC Interface and Output Mapping

Key points to cover.

- Inputs: qubit count, class count, feature subset, ansatz template, measurement mode, device and differentiation method.
- Materialisation of the PennyLane quantum layer and integration with PyTorch.
- Conversion of measured basis-state probabilities into class probabilities.
- Minimum versus full measurement mode.
- Training, prediction, scoring and result-extraction functions.

3.6.2 Feature Subspacing

- Selection of a subset of original features for each learner.
- Feature density as either a fraction or fixed number of features.
- Deterministic learner-specific seeds.
- Role in reducing circuit cost and increasing ensemble diversity.

Research notes and evidence.

- Random-subspace VQC studies report that smaller feature ratios can preserve performance while reducing training cost, although the optimum is dataset-dependent [17].
- Resource-saving QNN ensembles can reduce qubit requirements substantially, but lower feature coverage may also reduce predictive information [10].

3.6.3 Encoding Layers

- Angle-encoding layer and mapping of features to qubits.

- Feature strategies: same, shifted or explicit assignments.
- Feature re-uploading and feature count per qubit.
- Pairwise RZZ feature encoding where included.
- Barriers used only for circuit visualisation or transpilation clarity, if relevant.

3.6.4 Trainable and Entangling Layers

- Parameterised RX and RZ rotations.
- CZ and trainable RZZ entanglers.
- Linear, ring, cyclic, odd–even or other implemented connectivity patterns.
- Placement of entanglement relative to encoding and trainable layers.
- Parameter count and gate-count derivation.

3.7 VQC Architecture Exploration

3.7.1 Purpose

- Explain why the final ensemble cannot be evaluated fairly using one arbitrarily selected circuit.
- Identify the target: compact, trainable and sufficiently expressive base architectures rather than maximum standalone accuracy at any resource cost.

3.7.2 Search Space

- Qubit counts evaluated.
- Feature density and features per qubit.
- Minimum versus full measurement.
- Encoding style, feature strategy and re-upload count.
- Number and arrangement of trainable layers.
- Entangling-layer positions, connectivity and entangler type.
- Final exact values should be inserted from the executed job generator rather than memory.

3.7.3 Search Execution

- Job-list generation and stable job identifiers.

- Parallel execution on local or HPC resources.
- Early stopping, epoch budget and validation protocol.
- JSONL logging and later reduction of redundant output fields.
- Failed-run handling and result repair where needed.

3.7.4 Ranking and Selection

- Per-dataset ranking.
- Average rank or percentile across datasets.
- Overall configuration-level and ansatz-level aggregation.
- Performance consistency and resource cost.
- Shortlisting of qualitatively different high-performing base designs.
- Avoid selection using the final held-out test results.

Research notes and evidence.

- Ansatz performance can vary strongly even for circuits with similar depth and parameter count, justifying systematic architecture selection [16].
- Model-pool reuse separates expensive VQC training from later ensemble construction and analysis [16].
- Shallow architectures can provide strong F1-per-second behaviour and reduce noise exposure [16].

3.8 ECOC Base Ensemble

3.8.1 Coding Matrix Construction

- Define the implemented construction based on class count, learner count and ECOC depth.
- Explain how modular arithmetic generates learner-specific class groupings.
- Explain row/column orientation exactly as used in the implementation.
- Explain repetition or rolling when the requested learner count exceeds the initial matrix size.

3.8.2 Learner Initialisation

- One VQC per ECOC subtask.
- Learner-specific class count based on its code partition.

- Template assignment and feature-subset selection.
- Controlled use of same, few-different and all-different base architectures.

3.8.3 Training and Diversity

- Mapping original labels to each learner's ECOC targets.
- Stratified bagging or full-data training.
- Class weighting where used.
- Diversity from ECOC tasks, feature subsets, data samples, ansatz templates, entangler type and feature assignment.

3.8.4 Direct Measured Aggregation

- Independently measure each VQC.
- Convert learner probabilities to scores over original classes using the ECOC matrix.
- Describe weighted or unweighted aggregation.
- Decode the highest final class score.
- Establish this as the direct measured quantum ensemble baseline.

3.9 Classical Measured Stacking

3.9.1 Meta-Feature Construction

- Concatenate the measured probability outputs of all base VQCs.
- State the dimensionality of the resulting meta-feature vector.
- Explain whether original input features are included; use only the implemented final design.

3.9.2 Out-of-Fold Training Procedure

1. Split the training data into stratified folds.
2. Train a temporary ensemble on $K - 1$ folds.
3. Predict probabilities for the held-out fold.
4. Repeat until every training sample has an out-of-fold meta-feature vector.
5. Train the classical meta-learner on the complete out-of-fold matrix.
6. Train the final base ensemble on the full training data.
7. Generate final test meta-features with the final base ensemble.

3.9.3 Classical Meta-Learner

- State the final selected model, such as SVC or the implemented neural meta-learner.
- Include relevant hyperparameters and class weighting.
- Explain why this constitutes a strong measured baseline.

3.10 Quantum Re-Encoding Stacking

3.10.1 Measured Inputs

- Use the same base-output information available to classical stacking.
- Explain scaling, padding or feature selection applied before quantum encoding.

3.10.2 Meta-VQC

- Number of qubits and measured wires.
- Feature encoding of the base probability vector.
- Trainable and entangling layers.
- Loss, optimiser and training protocol.

3.10.3 Purpose of the Condition

- Isolate whether a quantum meta-model adds value when the information supplied to it is already classical.
- Separate the effect of quantum meta-model capacity from the effect of preserving base states.

3.11 Measurement-Free Quantum Stacking

3.11.1 Architectural Principle

- Each trained VQC becomes an unmeasured subcircuit within one larger quantum circuit.
- Base learners occupy disjoint registers and process their assigned feature subsets.
- No base-level probability measurement is performed.
- A quantum meta-circuit acts across selected wires or registers.
- Measurement occurs only at the final classifier output.

3.11.2 Base-Circuit Composition

- Reconstruct each selected ansatz inside the combined QNode.
- Load trained base parameters without copying the measured QLayer itself.
- Define the global wire mapping from local learner wires to the combined register.
- Preserve learner-specific feature subsets and circuit templates.

3.11.3 Frozen Base Parameters

- Train each base learner before constructing the coherent circuit.
- Freeze the base parameters during initial meta-training.
- Train only meta-level parameters to reduce optimisation difficulty.
- State whether end-to-end fine-tuning was attempted; omit it if not executed.

3.11.4 Meta-Wire Selection

- Main design: first or designated main wire from each base learner.
- Alternative design: all base wires, if evaluated.
- Explain the information–resource trade-off.

3.11.5 Cross-Learner Meta-Circuit

- Meta-level RX/RZ rotations.
- CZ or RZZ connections between learner registers.
- Number of meta-layers and connection topology.
- How cross-register entanglement differs from independent measured aggregation.

3.11.6 Final Measurement and Class Mapping

- Final measured wires.
- Basis-state to class-probability mapping.
- Treatment of unused basis states.
- Loss function used for final multiclass training.

3.11.7 Scaling Analysis

- Total width approximately equals the sum of base-register widths.
- Base-circuit depth remains parallel conceptually, but simulator and transpiled depth depend on scheduling and topology.
- Meta-depth grows with meta-layer count and connectivity.
- Two-qubit gate count grows with base entanglement plus cross-learner meta-entanglement.
- Statevector memory scales exponentially with total coherent qubit count.

Research notes and evidence.

- Staged optimisation of pretrained and frozen components is consistent with the broader hybrid variational approach and reduces the initial coherent optimisation burden [20, 8].
- Cross-register two-qubit gates introduce the mechanism of interest but also increase depth, topology overhead and hardware noise exposure [20].
- Multi-chip VQCs deliberately avoid inter-register entanglement for trainability, whereas this dissertation introduces a limited meta-circuit specifically to test coherent aggregation [15].

3.12 Classical Reference Models

Key points to cover.

- SVC configuration.
- Random Forest configuration.
- Preprocessing and data splits shared with quantum models.
- Explain that these provide practical context rather than perfectly capacity-matched quantum-advantage tests.

3.13 Training Protocol

3.13.1 Optimisation

- Adam or final selected optimiser.
- Learning rate.
- Negative log-likelihood or final selected loss.
- Batch size.
- Maximum epochs.

3.13.2 Early Stopping

- Warm-up period.
- Patience.
- Minimum loss improvement.
- Restoration of the best model state.

3.13.3 Repeated Training and Randomness

- Data-split seeds.
- Feature-subset and bagging seeds.
- Parameter initialisation.
- Number of repeated runs used in final comparisons.
- Explicitly acknowledge configurations with fewer repeats due to computational cost.

3.13.4 Execution and Failure Handling

- Multiprocessing and HPC worker allocation.
- Memory and qubit-count limits.
- Failed-run logging.
- Rules for retrying, excluding or reporting failed configurations.

3.14 Evaluation Framework

3.14.1 Predictive Metrics

- Accuracy.
- Macro-F1 as the primary class-balanced metric.
- Weighted-F1.
- Per-class precision and recall.
- Confusion matrices for selected comparisons.

3.14.2 Optimisation Metrics

- Best and final training loss.

- Best and final validation loss.
- Epochs completed and early-stopping epoch.
- Convergence curves for representative configurations.
- Run failures or unstable training.

3.14.3 Computational Metrics

- Base-training time.
- Out-of-fold stacking overhead.
- Meta-training time.
- End-to-end training time.
- Peak memory where available.

3.14.4 Quantum Resource Metrics

- Qubit count.
- Circuit depth.
- Trainable parameter count.
- Total gate count.
- Two-qubit gate count.
- Transpiled depth and SWAP overhead, if transpilation is performed.

Research notes and evidence.

- Accuracy alone can obscure minority-class failure, supporting macro-F1 and per-class recall [6, 28].
- Hardware feasibility depends on width, depth, two-qubit gate count, topology, gate fidelity and coherence rather than qubit count alone [20].
- Previous VQC ensemble studies evaluate performance and training cost jointly and use statistical tests where repeated observations permit them [17].

3.15 Noise and Hardware Evaluation

Key points to cover.

- Keep this section only if noisy simulation or hardware experiments are completed.

- Specify the noise model or exact backend and calibration date.
- Specify shot count, transpilation level and error-mitigation method.
- Use a reduced set of configurations selected before hardware testing.
- Compare degradation relative to noiseless simulation.
- Do not retain generic IBM hardware claims if the final work remains simulator-only.

Research notes and evidence.

- Ensemble QNNs and AdaBoost.Q have been tested on superconducting hardware, showing that some quantum ensemble methods can move beyond ideal simulation [10, 25].
- Hardware deployment requires topology-aware transpilation, qubit placement and error mitigation, and added two-qubit gates may remove a noiseless coherent advantage [19].

3.16 Reproducibility and Result Management

Key points to cover.

- Repository commit used for final experiments.
- Environment and dependency versions.
- Dataset download and preprocessing scripts.
- Configuration templates and job generator.
- Stable job IDs and skipped completed jobs.
- JSONL schema and reduced logging design.
- Aggregated parquet files and analysis dashboard.
- Random seeds and commands needed to reproduce the final tables.

Chapter 4: Evaluation and Results

4.1 Experimental Overview

Results to insert.

- Final datasets and their roles.
- Final model configurations and aggregation conditions.
- Number of runs, seeds and completed/failed jobs.
- Hardware and simulator resources used.
- Compact table of the final experimental matrix.

4.2 VQC Architecture Exploration

4.2.1 Search Coverage

- Number of generated and completed VQC jobs.
- Distribution across datasets, qubit counts and architecture groups.
- Exclusions caused by failure, memory limits or invalid configurations.

4.2.2 Overall Architecture Trends

- Effects of qubit count.
- Effects of feature density and features per qubit.
- Effects of re-uploading and trainable-layer count.
- Effects of entangler and entanglement placement.
- Effects of measurement mode.
- Report interactions only when supported by the analysis.

4.2.3 Dataset-Dependent Behaviour

- Which architectural choices generalise across datasets.

- Which choices are strongly dataset-specific.
- Relationship with class count, feature count and dataset difficulty.

4.2.4 Selected Base Configurations

- Final shortlist.
- Average percentile or ranking.
- Resource cost.
- Architectural diversity.
- Reason each design was retained for ensemble experiments.

4.3 Baseline Classification Performance

4.3.1 Classical Models

- SVC and Random Forest accuracy, macro-F1 and weighted-F1.
- Identify datasets where classical reference performance is already near ceiling.

4.3.2 Individual VQCs

- Performance distribution of selected VQCs.
- Variance across architecture and seed.
- Resource cost and training time.

4.3.3 Direct QuantumECOC

- Compare with individual VQCs.
- Evaluate the effect of ensemble size and base diversity.
- Report class-level gains or failures.
- Answer Research Question 1.

4.4 Classical Measured Stacking Results

4.4.1 Predictive Performance

- Direct QuantumECOC versus classical StackedECOC.
- Accuracy and macro-F1 differences by dataset.
- Mean/median difference across datasets and seeds.

4.4.2 Per-Class Behaviour

- Minority or difficult classes assisted by stacking.
- Confusion matrices for representative improvements and regressions.

4.4.3 Computational Cost

- Cost of K temporary fold ensembles.
- Final full-data retraining cost.
- Total stacking overhead relative to direct aggregation.

4.4.4 Research Question 2

- Give a direct evidence-based answer after the results are presented.

4.5 Quantum Re-Encoding Results

4.5.1 Predictive Performance

- Quantum re-encoding versus the classical meta-learner using the same measured base outputs.
- Dataset-level and aggregate comparisons.

4.5.2 Convergence and Stability

- Loss curves.
- Sensitivity to initialisation.
- Early-stopping behaviour.
- Failed or collapsed runs.

4.5.3 Research Question 3

- State whether making only the meta-model quantum provided a consistent benefit.

4.6 Measurement-Free Stacking Results

4.6.1 Implementation Feasibility

- Largest successfully simulated coherent circuit.
- Maximum learner and qubit counts.
- Memory, runtime and differentiation constraints.
- Configurations that could not be trained and the observed failure mode.
- Answer the feasibility component of Research Question 4.

4.6.2 Predictive Performance

- Coherent stacking versus direct QuantumECOC.
- Coherent stacking versus classical stacking.
- Coherent stacking versus quantum re-encoding.
- Accuracy, macro-F1 and per-class behaviour.

4.6.3 Convergence and Stability

- Training and validation curves.
- Effect of frozen base parameters.
- Meta-gradient and optimiser behaviour if analysed.
- Variability across seeds.

4.6.4 Effect of Ensemble Size

- Predictive effect of increasing learner count.
- Qubit, depth, gate and simulation-cost growth.
- Point at which added learners stop being practical or beneficial.

4.6.5 Effect of Meta-Circuit Design

- Main wires versus all wires, if evaluated.
- Meta-layer count.
- CZ versus RZZ.
- Entanglement topology.

4.6.6 Dataset-Dependent Effects

- Association with class count, base weakness, imbalance or feature complexity.
- Distinguish robust patterns from isolated best scores.

4.6.7 Research Questions 4 and 5

- Give direct answers on feasibility and predictive benefit.

4.7 Resource and Scalability Analysis

4.7.1 Circuit Width

- Formula and measured values by ensemble size.

4.7.2 Depth and Gate Counts

- Logical circuit depth.
- Total gates.
- Two-qubit gates.
- Transpiled values where available.

4.7.3 Training Time and Memory

- Runtime scaling by qubit and learner count.
- Base training, fold training and coherent meta-training.
- Peak memory or practical cluster limits.

4.7.4 Performance–Cost Trade-Off

- Performance gain per additional qubit, two-qubit gate or training hour.
- Identify configurations that are dominated by cheaper alternatives.
- Answer Research Question 6.

4.8 Noise or Hardware Results

Results to insert.

- Retain only if the corresponding methodology was executed.
- Noiseless versus noisy simulation.
- Simulator versus hardware.
- Effect of transpilation and mitigation.
- Whether any coherent advantage survives realistic noise.

4.9 Statistical and Practical Significance

Key points to cover.

- Confidence intervals or standard deviations across repeats.
- Paired comparisons using shared splits/seeds.
- Friedman/Wilcoxon or another justified test when enough observations exist.
- Effect sizes and consistency, not only p-values.
- Distinguish numerical best scores from practically meaningful improvements.

4.10 Summary by Research Question

Table 4.1: Summary of evidence and findings by research question.

RQ	Evidence	Finding
RQ1	Individual VQC and direct QuantumECOC results	*To complete*
RQ2	Direct aggregation versus classical stacking	*To complete*
RQ3	Classical meta-learner versus quantum re-encoding	*To complete*
RQ4	Coherent implementation and scaling observations	*To complete*
RQ5	Coherent versus measured predictive results	*To complete*
RQ6	Width, depth, gates, time and memory	*To complete*
RQ7	Dataset and ensemble characteristic analysis	*To complete*

Chapter 5: Discussion

5.1 Interpretation of the Main Findings

Key points to cover.

- Explain the overall pattern across direct aggregation, classical stacking, quantum re-encoding and coherent stacking.
- Identify whether performance was mainly determined by base-learner quality, learned aggregation, quantum meta-model capacity or state preservation.
- Interpret unsuccessful or unstable coherent experiments as evidence about feasibility rather than omitting them.

5.2 Is Measurement an Information Bottleneck?

Key points to cover.

- Return directly to the central hypothesis.
- Separate three possibilities: coherent improvement, no meaningful difference or worse coherent performance.
- Explain that poor coherent performance does not prove that the preserved state contains no useful information; optimisation, circuit design and resource limits may prevent its extraction.
- Avoid equating simulator state access with a practical hardware advantage.

5.3 When Does Stacking Help?

Key points to cover.

- Relate stacking gains to base-learner weakness.
- Examine class count, imbalance, feature dimensionality and dataset difficulty.
- Discuss whether simple datasets leave little room for a meta-learner.
- Discuss whether harder datasets benefit from learned aggregation but incur much higher training cost.
- Answer Research Question 7.

5.4 Quantum Meta-Learning versus Coherent Aggregation

Key points to cover.

- Contrast placing a quantum model after measurement with maintaining a continuous quantum computation across base and meta stages.
- Use the re-encoding condition to determine whether the meta-model type or the information boundary is more influential.
- Discuss capacity matching and optimisation fairness between classical and quantum stackers.

5.5 Performance–Cost Trade-Off

Key points to cover.

- Evaluate whether any coherent performance gain justifies the extra qubits, depth, gates, memory and runtime.
- Discuss whether a smaller measured model is practically preferable even when coherent stacking achieves a slightly higher score.
- Identify the most promising configuration under a balanced performance–resource criterion.

5.6 Relationship to Existing Literature

Key points to cover.

- Compare the observed ensemble gains with measured quantum bagging and boosting studies.
- Compare the classical stacking findings with measured quantum stacking literature.
- Compare the re-encoding findings with quantum meta-learning work.
- Compare coherent feasibility with theoretical or indexed coherent ensemble proposals.
- Explain which findings confirm, extend or conflict with earlier studies.

5.7 Limitations

Key points to cover.

- Small or moderate tabular datasets.
- Limited repeated runs for expensive configurations.
- Restricted hyperparameter search for the coherent condition.
- Exponential statevector simulation cost and resulting learner-count ceiling.
- Simulator-only conclusions if no hardware experiments are completed.
- Unequal representational capacity across classical and quantum meta-learners.
- Dependence on selected ansatz families, feature mapping and output measurement.
- Frozen-base training may prevent globally optimal end-to-end solutions.
- Accuracy and macro-F1 provide indirect rather than information-theoretic evidence about measurement loss.
- Internal student projects should be used as design guidance and clearly distinguished from peer-reviewed evidence [17, 16].

5.8 Threats to Validity

5.8.1 Internal Validity

- Data leakage, preprocessing differences, random initialisation, feature selection and unequal optimisation budgets.

5.8.2 Construct Validity

- Whether predictive metrics adequately represent preserved quantum information or information efficiency.

5.8.3 External Validity

- Generalisation from small tabular datasets and simulation to larger datasets and physical devices.

5.8.4 Conclusion Validity

- Number of repeats, failed runs, statistical power and sensitivity to selected datasets.

Chapter 6: Conclusion and Future Work

6.1 Dissertation Summary

Key points to cover.

- Restate the problem and motivation.
- Summarise the progression from VQC selection and ECOC construction to the three stacking conditions.
- Summarise the final experimental scope.

6.2 Answers to the Research Questions

Key points to cover.

- Answer each research question explicitly in order.
- Use concise evidence-based statements rather than repeating the whole results chapter.
- State uncertainty and conditional conclusions where the experiments are limited.

6.3 Contributions

Key points to cover.

- Restate only completed implementation contributions.
- State empirical findings separately from software or methodological contributions.
- Include informative negative results, practical scaling limits and failed approaches where they advance understanding.

6.4 Overall Conclusion

Key points to cover.

- Give the final judgement on whether measurement-free stacking was feasible.
- State whether it offered a consistent predictive advantage.
- State whether any gain justified its additional resource and optimisation cost.
- Avoid broader quantum-advantage claims beyond the evidence.

6.5 Future Work

Key points to cover.

- End-to-end fine-tuning after frozen-base meta-training.
- Alternative cross-register meta-circuits and learned connectivity.
- More direct diagnostics of phase, amplitude or mutual-information preservation.
- Larger coherent ensembles using tensor-network, distributed or approximate simulation.
- Hardware-aware training, topology-constrained meta-circuits and error mitigation.
- Real IBM hardware evaluation of a reduced coherent configuration.
- Stronger repeated-seed and statistical evaluation.
- Capacity-matched monolithic VQC and classical neural meta-learner comparisons.
- Automated selection based on prediction diversity rather than architecture labels alone.
- Release or formal evaluation of the VQC architecture-search pipeline as a secondary research tool.

Appendix A: **Dataset Details**

- Data sources and licences.
- Original and cleaned dimensions.
- Feature definitions and class distributions.
- Dataset-specific preprocessing.

Appendix B: VQC and Ansatz Configurations

- Layer specifications.
- Final template JSON definitions.
- Parameter and gate counts.
- Representative circuit diagrams.

Appendix C: Experimental Search Spaces

- VQC search grid.
- Ensemble composition grid.
- Meta-circuit grid.
- Final reduced stress-test configurations.

Appendix D: **Additional Results**

- Full per-dataset tables.
- Classification reports.
- Confusion matrices.
- Loss curves.
- Failed-run summaries.

Appendix E: Circuit and Resource Details

- Individual VQC circuit.
- Direct QuantumECOC information flow.
- Re-encoding meta-VQC.
- Measurement-free coherent circuit.
- Logical and transpiled resource tables.

Appendix F: Computational Environment and Reproducibility

- Software versions.
- HPC resources and Slurm scripts.
- Commands for job generation, execution and aggregation.
- Result schemas and dashboard instructions.

Acknowledgements

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